Quantum Scientific Computing (QSC2026)

Homework 1 — Solutions

These solutions show the intended level of detail; many admit equivalent routes. Numerical answers are exact where a closed form exists and rounded to four digits otherwise.

Problem 1 (Single qubit). $|\psi\rangle = a\,|0\rangle + b\,|1\rangle$ with $a = \cos\frac{\pi}{8}$, $b = \mathrm{e}^{\mathrm{i}\pi/4}\sin\frac{\pi}{8}$.

Solution. **(a)** $|a|^2 + |b|^2 = \cos^2\frac{\pi}{8} + \sin^2\frac{\pi}{8} = 1$. The computational-basis probabilities are $\mathbb{P}(0) = \cos^2\frac{\pi}{8} = \frac{2+\sqrt{2}}{4} \approx 0.8536$ and $\mathbb{P}(1) = \sin^2\frac{\pi}{8} = \frac{2-\sqrt{2}}{4} \approx 0.1464$.

(b) With $\bar{a}b = \cos\frac{\pi}{8}\sin\frac{\pi}{8}\,\mathrm{e}^{\mathrm{i}\pi/4} = \frac{1}{2}\sin\frac{\pi}{4}\,\mathrm{e}^{\mathrm{i}\pi/4} = \frac{\sqrt{2}}{4}\mathrm{e}^{\mathrm{i}\pi/4}$,

$$\langle X\rangle = 2\Re(\bar{a}b) = \tfrac{\sqrt{2}}{4}\cdot 2\cos\tfrac{\pi}{4} = \tfrac{1}{2}, \quad \langle Y\rangle = 2\Im(\bar{a}b) = \tfrac{1}{2}, \quad \langle Z\rangle = |a|^2 - |b|^2 = \cos\tfrac{\pi}{4} = \tfrac{\sqrt{2}}{2}.$$

Bloch vector $\mathbf{r} = (\frac{1}{2}, \frac{1}{2}, \frac{\sqrt{2}}{2})$, with $|\mathbf{r}|^2 = \frac{1}{4} + \frac{1}{4} + \frac{1}{2} = 1$ (pure state).

(c) $\mathbb{P}(+) = |\langle +|\psi\rangle|^2 = \frac{1}{2}\,|a+b|^2 = \frac{1}{2}(1 + 2\Re(\bar{a}b)) = \frac{1}{2}(1 + \langle X\rangle) = \frac{1}{2}(1+\frac{1}{2}) = \frac{3}{4}$. (The identity $\mathbb{P}(+) = \frac{1}{2}(1+\langle X\rangle)$ is the general rule for an X-basis measurement.)

(d) A Z-measurement returns ± 1 with variance $\mathrm{Var} = 1 - \langle Z\rangle^2 = 1 - \frac{1}{2} = \frac{1}{2}$; the empirical mean has error $\approx z\sqrt{\mathrm{Var}\,/N}$ at confidence level set by z. Setting this to $\epsilon = 0.05$ with $z \approx 2$ gives $N \approx z^2\,\mathrm{Var}\,/\epsilon^2 = 4\cdot 0.5/0.0025 \approx 800$. The point is the shot-noise law $N = \Theta(1/\epsilon^2)$ (hundreds to a few thousand shots here).

Problem 2 (Separability test). A state $a\,|00\rangle + b\,|01\rangle + c\,|10\rangle + d\,|11\rangle$ factorizes iff $ad = bc$.

Solution. **(a)** $a = b = c = d = \frac{1}{2}$: $ad = \frac{1}{4} = bc$, so it is a *product* state, $= \frac{1}{2}(|0\rangle + |1\rangle)(|0\rangle + |1\rangle) = |+\rangle \otimes |+\rangle$.

(b) $a = \frac{1}{\sqrt{2}}$, $d = -\frac{1}{\sqrt{2}}$, $b = c = 0$: $ad = -\frac{1}{2} \neq 0 = bc$, so it is *entangled* (the Bell state $|\Phi^-\rangle$).

(c) $a = b = c = \frac{1}{2}$, $d = -\frac{1}{2}$: $ad = -\frac{1}{4} \neq \frac{1}{4} = bc$, so it is *entangled*.

(d) If $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (\alpha\,|0\rangle + \beta\,|1\rangle)(\gamma\,|0\rangle + \delta\,|1\rangle)$ then matching coefficients gives $\alpha\gamma = \beta\delta = \frac{1}{\sqrt{2}}$ (so $\alpha, \beta, \gamma, \delta \neq 0$) but $\alpha\delta = \beta\gamma = 0$ (coefficients of $|01\rangle, |10\rangle$), a contradiction. Hence no factorization exists; equivalently $ad - bc = \frac{1}{2} \neq 0$.

Problem 3 (Gates, reversibility, no-cloning).

Solution. **(a)** $(H \otimes \mathbb{I})\,|00\rangle = |+\rangle\,|0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$; then CNOT (control first) sends $|10\rangle \mapsto |11\rangle$, giving $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = |\Phi^+\rangle$.

(b) Toffoli: $|a, b, c\rangle \mapsto |a, b,\, c \oplus (a \wedge b)\rangle$, so with $c = 0$ it writes $a \wedge b$ onto the third qubit. It is its own inverse, hence reversible. The bare map $(a, b) \mapsto a \wedge b$ is $2 \to 1$ and not injective (both 01 and 00 give 0), so it cannot be a unitary on two qubits; keeping a, b and writing the result on a fresh qubit restores invertibility.

(c) Take $U = \mathrm{CNOT}$ and $|s\rangle = |0\rangle$: then $U\,|0\rangle\,|0\rangle = |00\rangle$ and $U\,|1\rangle\,|0\rangle = |11\rangle$ as required. On a general input, $U(\alpha\,|0\rangle + \beta\,|1\rangle)\,|0\rangle = \alpha\,|00\rangle + \beta\,|11\rangle$, whereas a true copy would be $(\alpha\,|0\rangle + \beta\,|1\rangle)^{\otimes 2} = \alpha^2\,|00\rangle + \alpha\beta\,|01\rangle + \alpha\beta\,|10\rangle + \beta^2\,|11\rangle$. These agree only when $\alpha\beta = 0$, i.e. for basis inputs; U entangles instead of copying. Linearity forbids a universal cloner—this is the no-cloning theorem.

Problem 4 (Teleportation). Message $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ on qubit 1; shared $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ on qubits 2 (Alice) and 3 (Bob).

Solution. **(a)** The joint state $\frac{1}{\sqrt{2}}(\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle)$ re-expressed in the Bell basis of qubits $1, 2$ is

$$\tfrac{1}{2}\Big[|\Phi^+\rangle(\alpha|0\rangle + \beta|1\rangle) + |\Phi^-\rangle(\alpha|0\rangle - \beta|1\rangle) + |\Psi^+\rangle(\alpha|1\rangle + \beta|0\rangle) + |\Psi^-\rangle(\alpha|1\rangle - \beta|0\rangle)\Big]_3.$$

With the standard $\mathrm{CNOT}_{1\to 2}$ then H_1 measurement giving bits (m_1, m_2), the four outcomes and Bob's pre-correction states are $(0,0)\to|\psi\rangle$, $(0,1)\to X|\psi\rangle$, $(1,0)\to Z|\psi\rangle$, $(1,1)\to XZ|\psi\rangle$, each with probability $\frac{1}{4}$.

(b) For $(m_1, m_2) = (1,1)$ the pre-correction state is $XZ|\psi\rangle = \alpha|1\rangle - \beta|0\rangle$. Bob applies X then Z (i.e. $Z^{m_1}X^{m_2}$): $X(\alpha|1\rangle - \beta|0\rangle) = \alpha|0\rangle - \beta|1\rangle = Z|\psi\rangle$, then $Z(\alpha|0\rangle - \beta|1\rangle) = \alpha|0\rangle + \beta|1\rangle = |\psi\rangle$. Restored.

(c) Before the bits arrive, all four outcomes are equally likely, so Bob's density operator is $\frac{1}{4}\sum_{P\in\{\mathbb{I},X,Z,XZ\}} P|\psi\rangle\langle\psi|P^\dagger = \frac{1}{2}\mathbb{I}$, the maximally mixed state—independent of $|\psi\rangle$. Nothing Alice does changes Bob's local statistics until the two classical bits (sent no faster than light) arrive; hence no faster-than-light signaling.

Problem 5 (Phase kickback; Deutsch; Deutsch–Jozsa).

Solution. **(a)** $U_f|x\rangle\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}}|x\rangle(|f(x)\rangle - |1\oplus f(x)\rangle) = (-1)^{f(x)}|x\rangle\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = (-1)^{f(x)}|x\rangle|-\rangle$, since the pair $(|f(x)\rangle, |1\oplus f(x)\rangle)$ equals $(|0\rangle, |1\rangle)$ if $f(x) = 0$ and $(|1\rangle, |0\rangle)$ if $f(x) = 1$.

(b) The four functions: $f \equiv 0$ and $f \equiv 1$ are *constant*; $f(x) = x$ and $f(x) = 1 - x$ are *balanced*. Deutsch prepares $|+\rangle|-\rangle$; phase kickback gives the first qubit $\propto (-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle$. Applying H: if $f(0) = f(1)$ the qubit is $\pm|+\rangle \xrightarrow{H} \pm|0\rangle$ (measure 0); if $f(0) \neq f(1)$ it is $\pm|-\rangle \xrightarrow{H} \pm|1\rangle$ (measure 1). So the single query returns 0 iff f is constant.

(c) In Deutsch–Jozsa the amplitude of $|0^n\rangle$ is $2^{-n}\sum_x(-1)^{f(x)}$: it equals ± 1 (probability 1) when f is constant and 0 when f is balanced. The promise "f is constant or balanced" is what makes the one-query, all-zeros test exact.

Problem 6 (Grover search, $N = 16$).

Solution. **(a)** $\sin\frac{\theta}{2} = \frac{1}{\sqrt{16}} = \frac{1}{4}$, so $\theta = 2\arcsin\frac{1}{4} \approx 0.5054$ ($\approx 28.96°$). $R_{x_0} = \mathbb{I} - 2|x_0\rangle\langle x_0|$ reflects about the axis $|\psi_0^\perp\rangle$ and $R_{\psi_0} = 2|\psi_0\rangle\langle\psi_0| - \mathbb{I}$ reflects about $|\psi_0\rangle$; a product of two reflections is a rotation by twice the angle between their mirror axes. Since $|\psi_0\rangle$ sits at angle $\theta/2$ from $|\psi_0^\perp\rangle$, $G = R_{\psi_0}R_{x_0}$ rotates by θ toward $|x_0\rangle$.

(b) $p_k = \sin^2((2k+1)\theta/2)$:

$$p_0 = 0.0625,\ p_1 = 0.4727,\ p_2 = 0.9084,\ p_3 = 0.9613,\ p_4 = 0.5817,\ p_5 = 0.1255.$$

The optimum is $k^* = 3$ ($p_3 \approx 0.961$). At $k = 4, 5$ the angle $(2k+1)\theta/2$ passes $\frac{\pi}{2}$, so the state rotates *past* $|x_0\rangle$ and the success probability falls back (over-rotation).

(c) For $N = 2^n$ with one marked item, $\theta/2 = \arcsin(1/\sqrt{N}) \approx 1/\sqrt{N}$; maximizing wants $(2k+1)\theta/2 \approx \frac{\pi}{2}$, giving $k^* \approx \frac{\pi}{4}\sqrt{N}$ and query complexity $\mathcal{O}(\sqrt{N})$.

Problem 7 (Hadamard test).

Solution. **(a)** After H on the ancilla and controlled-U the state is $\frac{1}{\sqrt{2}}(|0\rangle|\psi\rangle + |1\rangle U|\psi\rangle)$; a second H gives $\frac{1}{2}[|0\rangle(|\psi\rangle + U|\psi\rangle) + |1\rangle(|\psi\rangle - U|\psi\rangle)]$, so $p(0) = \frac{1}{4}\||\psi\rangle + U|\psi\rangle\|^2 = \frac{1}{4}(2 + 2\Re\langle\psi|U|\psi\rangle) =$

$\frac{1}{2}(1+\Re\langle\psi|U|\psi\rangle)$. Inserting $S^\dagger=\mathrm{diag}(1,-\mathrm{i})$ on the ancilla between the two Hadamards replaces $\Re$ by $\Im$: $p(0)=\frac{1}{2}(1+\Im\langle\psi|U|\psi\rangle)$.

(b) $U=S=\mathrm{diag}(1,\mathrm{i})$, $|\psi\rangle=|+\rangle$: $\langle+|S|+\rangle=\frac{1}{2}(\langle0|+\langle1|)(|0\rangle+\mathrm{i}\,|1\rangle)=\frac{1}{2}(1+\mathrm{i})$, so $\Re=\Im=\frac{1}{2}$. The real-part test predicts $p(0)=\frac{1}{2}(1+\frac{1}{2})=\frac{3}{4}$ and the imaginary-part variant also $\frac{3}{4}$, consistent with $\langle+|S|+\rangle=(1+\mathrm{i})/2$.

Problem 8 (QFT and phase estimation).

Solution. **(a)** For $n=3$, $\mathrm{QFT}\,|j\rangle=\frac{1}{\sqrt{8}}(|0\rangle+\mathrm{e}^{2\pi\mathrm{i}\,0.j_3}\,|1\rangle)(|0\rangle+\mathrm{e}^{2\pi\mathrm{i}\,0.j_2j_3}\,|1\rangle)(|0\rangle+\mathrm{e}^{2\pi\mathrm{i}\,0.j_1j_2j_3}\,|1\rangle)$. The circuit uses 3 Hadamards and $\binom{3}{2}=3$ controlled-phase gates (R_2,R_3), plus the final bit-reversal swap.

(b) $\varphi=\frac{3}{8}$, $t=3$. After the controlled powers the ancilla is $\frac{1}{\sqrt{8}}\sum_{j=0}^{7}\mathrm{e}^{2\pi\mathrm{i}(3/8)j}\,|j\rangle=\mathrm{QFT}\,|3\rangle$, because $\varphi=k/8$ with $k=3$. Applying QFT^{-1} returns $|3\rangle=|011\rangle$, measured with probability 1. The readout is *exact* because $\varphi=k/2^t$ is a dyadic rational with a t-bit expansion.

(c) Using $t=n+\lceil\log_2(2+\frac{1}{2\delta})\rceil$ for success $\geq 1-\delta$: with $n=6$, $\delta=0.1$ we get $\lceil\log_2(2+5)\rceil=\lceil\log_2 7\rceil=3$, hence $t=9$ ancilla qubits.

Problem 9 (Ground-state energy estimation).

Solution. **(a)** $U\,|E_j\rangle=\mathrm{e}^{-\mathrm{i}Ht}\,|E_j\rangle=\mathrm{e}^{-\mathrm{i}E_jt}\,|E_j\rangle=\mathrm{e}^{2\pi\mathrm{i}\varphi_j}\,|E_j\rangle$ with $\varphi_j=-\frac{E_jt}{2\pi}\bmod 1$. QPE estimates φ_j; inverting, $E_j=-\frac{2\pi}{t}\varphi_j$ (taken in the window chosen so $E_jt/2\pi\in[0,1)$).

(b) By linearity QPE on $|\phi\rangle=\sum_j c_j\,|E_j\rangle$ collapses onto $|E_j\rangle$ (returning φ_j) with probability $|c_j|^2$; the ground phase φ_0 thus appears with probability $p_0=|c_0|^2=0.4$ per run. For R independent runs, $\mathbb{P}(\text{seen at least once})=1-(1-p_0)^R\geq 0.99$ requires $(0.6)^R\leq 0.01$, i.e. $R\geq\ln 0.01/\ln 0.6\approx 9.02$, so $R=10$ runs suffice.

(c) Energy estimation only reads *some* eigenvalue each run and keeps the smallest observed; it needs the ground state to have nonnegligible overlap, but *not* a spectral gap, since one is not trying to resolve neighboring levels. Preparing the ground state (filtering out the excited components) does need a gap to separate E_0 from the rest.

Problem 10 (Amplitude estimation).

Solution. **(a)** Write $A\,|0^n\rangle=\sin\frac{\theta}{2}\,|\phi_{\mathrm{good}}\rangle+\cos\frac{\theta}{2}\,|\phi_{\mathrm{bad}}\rangle$, so $P=\sin^2\frac{\theta}{2}$. On the plane $\mathrm{span}\{|\phi_{\mathrm{good}}\rangle,|\phi_{\mathrm{bad}}\rangle\}$ the iterate $G=R_{\phi_0}R_{\mathrm{good}}$ is a rotation by θ with eigenvalues $\mathrm{e}^{\pm\mathrm{i}\theta}$; the prepared state is an equal superposition of the two eigenvectors, so QPE on G returns $\pm\theta$ and hence $P=\sin^2(\theta/2)$.

(b) For $\epsilon=10^{-3}$: amplitude estimation uses $\mathcal{O}(1/\epsilon)\approx 10^3$ applications of A (and of G), whereas classical Monte Carlo needs $\mathcal{O}(1/\epsilon^2)\approx 10^6$ samples—a factor $\approx 10^3$ fewer queries.

(c) The Hadamard test encodes $\Re\langle\psi|U|\psi\rangle$ in the bias $p(0)=\frac{1}{2}(1+\Re\langle\psi|U|\psi\rangle)$. Estimating $p(0)$ by repetition costs $\mathcal{O}(1/\epsilon^2)$ runs (shot noise); running amplitude estimation on the very same circuit estimates $p(0)$—hence $\Re\langle\psi|U|\psi\rangle$—to precision ϵ with $\mathcal{O}(1/\epsilon)$ coherent applications. This is the quadratic (Heisenberg-limited) improvement over plain repetition.